

NWFL Data Infrastructure & Production Roadmap

Prepared as the next operational phase for the NWFL data platform buildout.

1. Current Progress

- Match results + goals synced from Google Sheets ■
- AI parsers for WhatsApp text, images, and bulk match entry ■
- Basic analytics endpoints (goal distribution, game state) ■

2. Current Data Gaps

Data Type	Status	Why It Matters
Player Profiles	Missing	Goal scorers are stored only as strings. No player stats, bios, positions, or team affiliations.
Live Match Events	Missing	No substitutions, cards, halftime scores, possession, shots, or in-game trends.
Fixtures / Schedule	Missing	All 273 matches are final results. No upcoming games makes the platform less useful.
Standings Auto-Calculation	Partial	Standings table exists but is not automatically recalculated from match results.
Multi-Season Data	Missing	Only the 2024/25 season exists. No historical comparison capability.
Venue Data	Partial	Venue is only a text string with no stadium metadata, capacity, or location.
Referee Data	Missing	Officials are not tracked.
Transfer / Loan Data	Missing	No player movement tracking.
Injury / Suspension	Missing	No player availability tracking.

3. AI Infrastructure Reality Check

- **No web crawling:** Current AI only parses text manually provided by users. It cannot browse X, Facebook, YouTube, or Google in real time.
- **Free model limitations:** Current free-tier models have rate limits, no SLA guarantees, inconsistent output formats, and unreliable structured output.
- **No validation layer:** Hallucinated scores or player names can be written directly into the database.
- **No retry/fallback logic:** Single API failures can break sync operations.

4. Production-Grade Roadmap

Phase 1 — Foundation Stabilization (2–3 Weeks)

- Auto-recalculate standings from match results after every sync
- Add fixtures endpoint separating upcoming vs completed matches
- Create proper player model with persistent player records and statistics
- Create match events model for substitutions, cards, and halftime data
- Add validation layer before AI outputs are written to the database

Phase 2 — Data Ingestion Expansion (1–2 Months)

- Build scrapers for NWFL official site, sports blogs, and media outlets
- Integrate X (Twitter) API for live club updates
- Integrate YouTube Data API for highlights and goal compilations
- Integrate Facebook Graph API for club page updates
- Aggregate RSS feeds from sports publishers
- Add computer vision pipeline for extracting scoreboards from images/videos

Phase 3 — Monetization & Platform Expansion (2–3 Months)

- Launch public API as a paid data product
- Build embeddable widgets for blogs and news sites
- Offer premium analytics, predictions, and fantasy integrations
- Develop white-label admin infrastructure for other leagues

5. Strategic Direction

The immediate priority is stabilizing and structuring the data layer before expanding AI ingestion capabilities or monetization features. Without strong validation, proper schemas, and automated recalculation systems, scaling data ingestion will increase inconsistency and technical debt. The focus should therefore remain on: Reliable structured data models Automated standings and fixture systems Player and event-level tracking Validation and sync reliability Once the foundation is stable, the platform can evolve into a scalable football data infrastructure product for the NWFL ecosystem.